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FEATURE

Using ChatGPT in the Secondary Gifted Classroom for Personalized Learning and Mentoring

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Abstract: Artificial intelligence (AI) has the potential to revolutionize education by promoting personalized learning experiences and enhancing educators' efficiency. Generative AI tools like ChatGPT are increasingly used by K-12 teachers for lesson planning, differentiated instruction, and automated feedback. This article examines how ChatGPT, in alignment with the National Association for Gifted Children (NAGC) standards, can be used to address the unique cognitive and affective needs of secondary gifted learners. Case studies provide examples of how to use ChatGPT to facilitate differentiation, personalized learning, support independent research, and provide virtual mentorship. The importance of prompt engineering is emphasized and encouragement of critical thinking skills when using AI tools. By integrating human expertise with AI capabilities, educators can enhance the mentorship experience, foster reflective learning, and cultivate future-ready leaders.

Keywords: gifted education, ChatGPT, mentoring, personalized learning

Artificial intelligence (AI) experts have hyped its potential to revolutionize education by promoting more personalized learning experiences and significantly enhancing educators' efficiency. Generative AI tools (e.g., ChatGPT, Copilot, Gemini, Claude 3.5) have been leveraged by one-third of K-12 teachers in their classrooms to help design and modify lesson plans, differentiate instruction, automate grading and feedback, and assist in planning professional learning experiences (Langreo, 2024). While some critics have voiced concern over issues like plagiarism, bias, and data privacy, others suggest that with thoughtful implementation and guidance, it has the potential to transform the way we teach and learn (Ng, 2023). AI offers educators that work with students with gifts and talents new ways to tailor educational experiences to the unique cognitive and affective needs of their students (Siegle, 2023).

> "Educators are encouraged to explore and integrate ChatGPT into educational practices to effectively support gifted learners."

Gifted education programs provide instruction to support advanced learners' cognitive needs and their social and emotional development. In fact, the NAGC PreK-Grade 12 Gifted Programming Standards (National Association for Gifted Children, 2019) highlight student outcomes and evidence-based practices that address social-emotional areas across all six standards. Some of these needs include earlier career exploration and guidance due to students' multi-potentiality and diverse range of interests; mentoring; goal-setting and identification of talent development pathways to reach these goals; cultural competence; and development of

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their leadership and communication skills. AI can be used to address the additional needs of gifted and talented students from rural communities, diverse cultural or linguistic backgrounds, and those who are twice exceptional. In this article we will share with you a few examples of ways to use generative AI tools like ChatGPT to address NAGC standards related to guidance, mentoring, and advanced learning needs of students with gifts and talents.

What is Generative AI?

Artificial intelligence, or "machine-learning models that can learn to make predictions based on data" (Zewe, 2023, p. 2) has long been used in schools with instruments like self-adaptive assessments and educational computer games. At the end of 2022, a type of AI became widely available to the public when OpenAI released ChatGPT (https://chatgpt.com), now in its GPT-4o iteration and offering a GPT-4o mini version with knowledge through 2023. This generative artificial intelligence (GenAI) is a different form of machine-learning model that is trained to "create new data, rather than making a prediction about a specific dataset" (Zewe, 2023, p. 2). There are now numerous useful GenAI tools that can help educators customize lesson plans, differentiate activities, create presentations, grade student work and provide feedback, and modify instruction (see Table 1 for examples.)

One of the most popular GenAI tools, ChatGPT, is an advanced artificial intelligence (AI) system known as a large language model (LLM) that is designed to understand and generate human-like text based on the input it receives. Interacting with ChatGPT is like having conversations with an expert who can adapt their language and explanations to match the user's level of understanding. The architecture of GenAI provides adaptability with reflection and self-correction ability, which is crucial for gifted students who often grapple with complex concepts beyond their years. In the evolving educational landscape, AI tools like ChatGPT are invaluable for gifted learners, offering personalized learning, differentiation, a guide for independent research projects, and virtual mentoring. ChatGPT (when prompted with specific and targeted requests) acts as a knowledgeable mentor to meet the advanced learning needs of gifted students by providing differentiation that allows exploration beyond the standard curriculum.

Student Safety and Data Privacy

With the rise of AI use in education, policymakers and educators are focusing on children's rights and safety. An important consideration when planning instruction with students is data privacy. The Children's Online Privacy Protection Act (The Child Online Privacy Protection Act, 15 U.S.C. §§ 6501-6506) is a federal law that protects the privacy of children under age 13 online. It prohibits websites, online services, mobile apps, and social media companies from collecting individually identifiable information about children under age 13 without prior parental consent.

Several international organizations created frameworks for guidance on the use of AI with children, including the Institute of Electrical and Electronics Engineers (IEEE), United Nations International Children's Emergency Fund (UNICEF), and the World Economic Forum. UNICEF Innovation Human Rights Center UC Berkeley (2019) acknowledged the potential benefits of AI in education such as "personalized learning, helping kids develop social skills, enabling distance education for children in remote regions" (p. 11); however, they also highlight potential risks to human rights including "the right to an education, as well as the right to protection from exploitation and abuse, and the protection of children with disabilities" (p. 11).

Companies that use AI technology must disclose their privacy and data collection practices. In fact, both Google and Microsoft recently established principles of the ethical use of AI yet neither have shared policies specific to children (UNICEF Innovation Human Rights Center UC Berkeley, 2019). In light of these potential issues, educators and administrators should carefully review adaptive technology and generative AI tools' data protection and security policies before purchasing products and subscriptions. Furthermore, district policies must address parent consent for students between the ages of 13 and 18 and students of all ages should be engaged in conversation around the benefits and potential drawbacks of using AI tools. Consent forms should include information on how the tool will be used in class and how. Teachers working with students younger than 13 should use teacher-facing versus student-facing AI technology (i.e., the student does not use the AI directly; the teacher collects student input and uses the tool to create customized assignments) and know to maintain confidentiality when using GenAI products to individualize instruction, such as avoiding use of specific student data or personal information. Keeping these ideas in mind, let us explore the possibilities of GenAI through three cases of middle and high school gifted learners: Sophia, Mateo, and Kiara.[1]

Sophia: AI for Differentiation

One useful application of ChatGPT is its ability to provide differentiated, personalized learning experiences. Differentiated instruction is a method used to adapt to students' varying levels of readiness to learn, ability, and interests (Tomlinson, 2014). This is a core tenant of the NAGC Standards and encompasses

Table 1. AI Tools for Teachers.

Tool and Url	Description	Benefits	Limitations	Data Use Policy
Claude 3.5 Sonnet https://claude.ai/	AI powered natural language model for interactive text generation	Strong vision model can interpret graphs and charts; can generate code, text documents, websites	Lacks some useful features that ChatGPT-4o has, like memory/custom instructions	User inputs and outputs are not used to train the models
Goblin Tools https://goblin.tools/	Streamline tasks with specialized tools for neurodivergent individuals	Free; great for 2e and neurodivergent students	Limited tool selection; some users may not find interface intuitive	Does not store data on their servers and does not share it with any other third party
Hello History https://www.hellohistory.ai	Interactive tools for exploring historical knowledge through role-playing	Immersive historical contexts for students; experiential learning	Cost may be a barrier; requires devices with Internet access	Does not use student data to train AI models. All data is encrypted and protected
Knowji https://www.knowji.com/	Vocabulary learning through interactive and adaptive apps	Helpful for English language learners; adaptive learning features	Requires consistent use to be effective; limited to vocabulary learning	May use or share aggregate or anonymous data collected for purposes such as understanding or improving the service
MagicSchoolAI https://www.magicschool.ai	Assists teachers with developing lesson plans, designing assignments, generating materials, creating newsletters	Free version available; saves time with auto-generation of content aligned to standards; helps rewrite content for different reading levels; tools for different subject areas	Information not past 2021; may produce biased or incorrect content	See their student data policy that outlines specific guidelines for agreements with educational institutions: https://www.magicschool.ai/privacy-security/student-data-policy
Synthesia https://www.synthesia.io/	Transform text into videos featuring human presenters	Enhances visual and auditory learning experiences; high-quality video production	Cost may be prohibitive; video accuracy is dependent on text quality	Collects information such as use, feedback, and technical issues; Customer data are collected but users may request erasure of personal data

Standard 3: *Curriculum Planning and Instruction*. ChatGPT supports teachers in designing tiered assignments, creating extension activities, producing project-based learning opportunities, and generating challenging questions that promote critical thinking and creativity for gifted learners. Additionally, ChatGPT can help teachers quickly develop materials that are engaging and rigorous and cater to students' unique intellectual and social-emotional needs. By offering insights and resources that align with students' individual profiles, ChatGPT acts as a flexible tool that enables teachers to provide a more meaningful, personalized

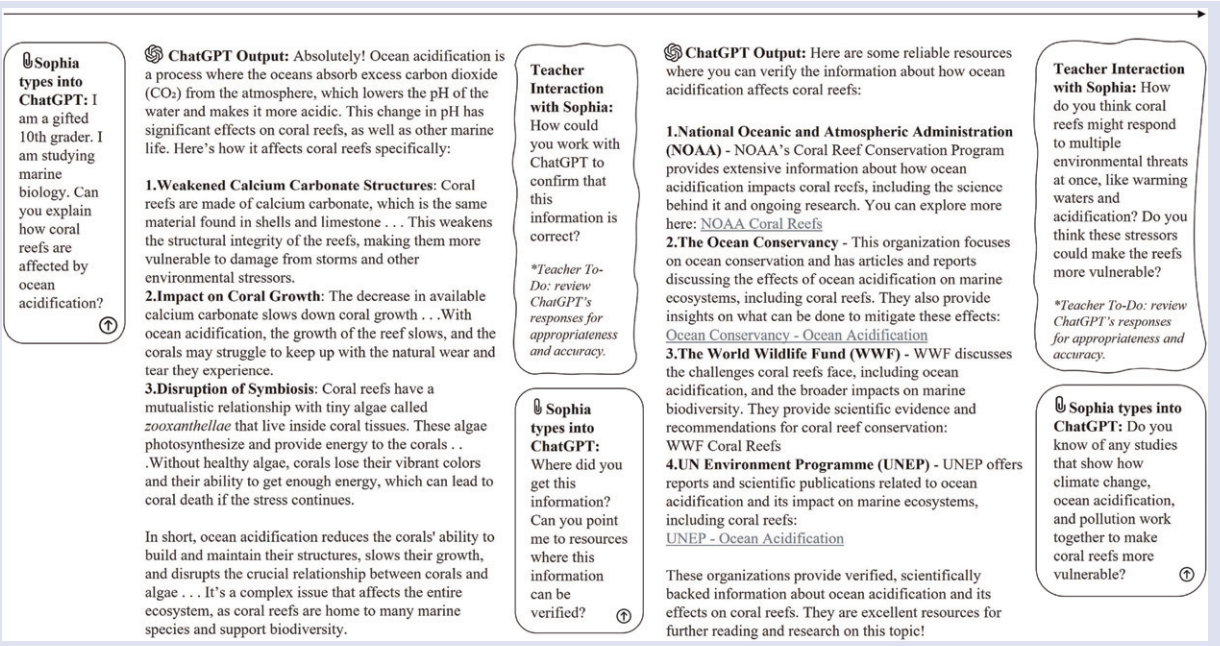


Figure 1. Critical thinking activities and reflection questions.

educational experience for each student. Whether modifying the complexity of tasks, providing enrichment opportunities, or helping to adjust pacing, ChatGPT serves as an accessible tool for enhancing the differentiation process.

Imagine a gifted high school student, Sophia, who lives in a rural community and lacks access to advanced science courses in her high school. Sophia's science teacher, Mr. Edwards, recognizes her passion for marine biology and wants to use ChatGPT to differentiate instruction so that she remains challenged and engaged. After obtaining a signed parental consent form (required for ChatGPT users between the ages of 13 and 18), Mr. Edwards adapts a standard grade-level science assignment on coral reefs into a more advanced, inquiry-driven project for Sophia. Instead of a typical research report, Sophia's assignment is restructured to include the exploration of deeper, more complex issues, such as the specific threats to coral reefs and the broader implications for marine biodiversity. As Sophia types more specific queries like "Can you explain how coral reefs are affected by ocean acidification?" ChatGPT tailors its responses by providing scientifically accurate, yet more detailed information to expand the complexity and depth of the topic. ChatGPT's adaptability allows Mr. Edwards to offer Sophia targeted resources, ask higher-level probing questions, and guide her toward advanced academic sources, such as NOAA and WWF, to help verify her findings as shown in Figure 1.

Through this modification, Sophia is encouraged not only to learn about coral reefs but also to engage in critical thinking by verifying the information ChatGPT provides. Mr Edwards also prompts her to reflect on the sources of information and encourages the use of additional credible resources, such as academic articles or conservation reports. As Sophia poses increasingly complex questions about deep-sea ecosystems, ChatGPT adeptly adjusts its responses, matching her level of understanding and pushing her knowledge boundaries further. This adaptive capability stems from ChatGPT's ability to analyze the nuances in Sophia's language, the complexity of her queries, and the context of their ongoing conversation. NAGC Standard 3, evidence-based practice 3.1.4. states, "Educators design differentiated curriculum that incorporates advanced, conceptually challenging, in-depth, and complex content for students with gifts and talents" (National Association for Gifted Children, 2019, p. 10). In this scenario, ChatGPT becomes a tool that assists Sophia's teacher in utilizing best practices for instruction and technology integration.

For educators to ensure ChatGPT-facilitated personalized learning is both tailored and accurate lies in guiding students like Sophia to provide clear context in their prompts and encouraging them to ask follow-up questions. For instance, Sophia might continue her session by stating, "How might the combined effects of climate change, ocean acidification, and pollution impact communities that rely on coral reefs, and what could they do to adapt?" This level of specificity helps ChatGPT calibrate its responses appropriately (see Table 2 for sample prompts).

Additionally, AI-assisted personalized learning requires educator oversight. According to Guilbault & Burrows (2023), "When ChatGPT gives an answer, it

Table 2. Sample ChatGPT Prompts—Sophia's Marine Biology Scenario.

Task	Examples
Student scenario prompts	Broad: "What are the most interesting things about the ocean?"
	Specific: "Can you tell me about different types of sea creatures in coral reefs?"
	Focused: "How do coral reefs work? What lives there?"
	Detailed: "Why do clownfish and sea anemones have a symbiotic relationship? How do they help each other?"
	Application: "How might the combined effects of climate change, ocean acidification, and pollution impact communities that rely on coral reefs, and what could they do to adapt?"
	Reflection: "If you were a marine biologist, what would you do to help protect coral reefs?"
Student final comprehensive prompt	"Imagine you're a marine biologist exploring a coral reef. Describe your adventure, including
	- Five different sea creatures you see and what they're doing
	- How the clownfish and sea anemones work together
	- Three ways the coral reef might be in danger
	- Your ideas to help protect the reef
	Be creative and use lots of details."
Teacher to-dos	1. Review ChatGPT's responses for reading level-appropriateness and accuracy
	2. Create a fact-checking activity: Have Sophia verify three interesting facts from ChatGPT's response using reliable sources
	3. Design a reflection exercise: "What surprised you most about coral reefs? What more do you want to learn?"
	4. Use ChatGPT's response as a basis for a creative writing or art project about ocean conservation

appears to be valid and trustworthy but is oftentimes not completely correct. This provides a prime opportunity for students to practice and develop their critical analysis skills" (p. 1). Hence, educators play a crucial role in regularly verifying ChatGPT's responses against standard curricula and expert knowledge, ensuring that the information students receive is not just engaging but also academically sound. One imperative strategy is to instruct students about the concept of AI hallucinations, where the model may generate incorrect or nonsensical information. This is a critical thinking skill for advanced learners that should be incorporated into any AI-assisted learning activities. Assignments should be designed with a critical thinking step included in which the learner evaluates the results from ChatGPT and confirms their validity. For example, Sophia's teacher could pose the following questions: (a) What new information did you learn from ChatGPT? (b) How did you verify the accuracy of this information? (c) What discrepancies did you find between ChatGPT's responses and other sources? and (d)

ChatGPT Critical Thinking Activities			
Activity	**Purpose**	**ChatGPT Follow-up Prompts**	**Student Reflection Questions**
Cross-Verification	Encourage students to ask ChatGPT to verify its own information or compare it with other sources. After students receive an answer, get into the habit of asking for a follow-up prompt.	"Can you provide sources or references for this information?"	"How does this information compare with what is found in [specific textbook or reliable website]?"
Error Identification	Train students to recognize potential errors. Help students to understand that the standard operating procedure when using ChatGPT is to check for errors.	"Please explain any possible limitations or errors in your previous response."	"What can I do to follow-up on the limitations or errors that ChatGPT shared?"
Extension Activities to Facilitate Fact-Checking and Reflection Skill Building			
Activity	**Directions**		
Reflective Journal	Have students keep journals to document the information they receive, their initial thoughts, and steps they took to verify the information.		
Group Discussions	Facilitate group discussions where students share what they have learned from ChatGPT and collaboratively evaluate the accuracy and reliability of the information. Students' critical analysis skills will be reinforced by the peer review process.		
Fact-Checking Assignments	Create assignments where students must fact-check responses from ChatGPT against multiple reliable sources. Students are tasked to research a specific topic and compare ChatGPT's information with academic articles, textbooks, and trusted websites.		
AI Think-Pair-Share Activity	Individual students craft their response to a question, then pair up with another student to compare their responses to how ChatGPT responds. Whose response is more accurate? Why? Working together the pair can evaluate and critique all three responses, synthesize, and share with the whole class.		

Figure 2. Critical thinking activities and reflection questions.

Table 3. Sample ChatGPT Prompts—Mateo's French Revolution Research.

Task	Examples
Student scenario prompts	Broad: "What was the French revolution all about?"
	Specific: "Why were people in France unhappy before the revolution?"
	Focused: "How was life different for rich and poor people in France back then?"
	Detailed: "Can you explain the tax system in pre-revolutionary France like I'm a child?"
	Application: "If you were a poor farmer in France during this time, how might your life change after the revolution?"
	Reflection: "What lessons do you think we can learn from the French Revolution that are still important today?"
Student final comprehensive prompt	"Pretend you're a time-traveling reporter visiting France just before the Revolution. Write a news article that includes
	- A catchy headline
	- What daily life is like for a rich person and a poor person
	- Three reasons why people are upset
	- An interview with a French citizen (you choose who!)
	- A prediction about what might happen next
	Use basic vocabulary, short sentences, and include a small glossary of five important terms."
Teacher to-dos	1. Use ChatGPT to generate a list of key vocabulary in both English and Mateo's first language
	2. Create a comparison activity: Have Mateo identify similarities and differences between ChatGPT's description and a textbook account using a mind map
	3. Design a critical thinking exercise: "If you were the King of France then, what could you do to help people?"
	4. Use ChatGPT's response to create a simple role-playing scenario for the class

How would you rate the reliability of the information provided, and why? Figure 2 provides sample activities to use with ChatGPT to encourage critical thinking and reflection.

Mateo: AI as an Independent Research Guide

NAGC Standard 3, student outcome 3.5 *Instructional Strategies* states, "Students with gifts and talents become independent investigators" (National Association for Gifted Children, 2019, p. 11). AI can be used to help educators address this outcome. One of ChatGPT's most powerful features is its ability to provide instant access to a vast reservoir of information, a perfect fit for gifted students with insatiable curiosity. Consider Mateo, a gifted English language learner with a passion for history. In mere seconds, ChatGPT provides Mateo with

a comprehensive overview of the socio-economic factors leading to the French Revolution, complete with key dates, figures, and events. The power behind this instant access lies in GenAI's ability to quickly synthesize information on a wide range of topics. It achieves this through its vast training data and sophisticated natural language processing algorithms. When Mateo asks about the French Revolution, ChatGPT doesn't simply retrieve a pre-written paragraph. Instead, it dynamically generates a response by drawing connections between relevant pieces of information in its knowledge base. ChatGPT nurtures Mateo's curiosity by encouraging and responding to his open-ended questions. His teacher combines the use of ChatGPT with Knowji, a GenAI app that helps build vocabulary (see Table 1), to help Mateo with any new sophisticated vocabulary that he is unfamiliar with, thus strengthening his English language skills and content expertise.

To ensure accuracy, again, it's crucial to instruct students like Mateo to frame specific, well-structured questions (see Table 3). For instance, instead of asking "Tell me about the French Revolution," Mateo could ask, "What were the primary economic factors that contributed to the onset of the French Revolution, and how did they interact with social and political issues of the time?" This specificity guides ChatGPT to provide more focused and relevant information. However, while it's important to use ChatGPT as a starting point for deeper research, it is by no means a final authority. While it provides quick, synthesized information, in-depth understanding often requires more extensive study. Mateo could use ChatGPT's overview as a foundation to identify areas for further exploration, formulate more specific questions, or discover new avenues of historical inquiry.

Kiara: AI as a Virtual Mentor

Kiara is a twice exceptional student who is interested in astrophysics and whose role model is Neil deGrasse Tyson, PhD. In a traditional classroom, she might struggle to find resources that challenge her beyond the standard science curriculum due to her asynchronous or uneven development (Guilbault & Kane, 2016). With ChatGPT, Kiara can dive into discussions with her hero, Dr deGrasse Tyson, about black holes, careers in quantum mechanics, and the nature of dark matter, receiving explanations tailored to her advanced understanding but still accessible for her age. By asking ChatGPT, "Pretend you are Neil deGrasse Tyson. Explain to me how you believe black holes work," she is instantly having a thought-provoking discussion with her virtual mentor. For other ideas of how to use ChatGPT to have students access the voices of historical figures and authors see https://www.hellohistory.ai. Her teacher, Mrs. Spencer, might also use MagicToDo (https://www.goblin.tools) to help Kiara break down her project into smaller, manageable chunks that keep her organized.

Maximizing the true power of ChatGPT requires the adoption and mastery of the art of prompt engineering. Educators and students need to learn how to craft questions and instructions that elicit the most useful and insightful responses from the AI model. It's like learning to speak a new language—the language of AI interaction. Effective prompt engineering is akin to being a skilled interviewer. Instead of asking ChatGPT to simply "explain black holes," Kiara might learn to phrase her query more precisely: "Explain the concept of Hawking radiation in black holes, comparing it to similar phenomena in quantum field theory. Use analogies suitable for a high school physics student." This level of specificity guides ChatGPT to provide a response that is not only informative but also tailored to Kiara's advanced level of understanding (see Table 4 for prompt examples).

Using GenAI for Guidance and Future Planning

Gifted students may have areas of interest and expertise that appear at a young age. They often require career exploration and guidance at an earlier age than typical, especially if they experience multi-potentiality. NAGC Standard 1, *Learning and Development*, recommends educators use evidence-based practices that help students with gifts and talents identify their interests, strengths, and needs and gather information to "inform students and families about resources available to develop their child's talents" (National Association for Gifted Children, 2019, p. 6) and "help students identify college and career goals that are consistent with their interests and strengths" (2019, p. 7). In addition, Standard 5, *Programming* includes student outcome 5.3 related to career pathways that states, "Students with gifts and talents create future career-oriented goals and identify talent development pathways to reach those goals" (2019, p. 15). Let's explore a few ways teachers can utilize GenAI to address these objectives.

Creating Individualized Learning Plans

As ChatGPT becomes one of Sophia, Mateo, and Kiara's constant companions on their learning journey, teachers can use it for guidance on various educational paths and assistance with decision-making processes. To effectively tailor the learning experience, formal assessments (e.g., standardized tests, unit tests, and grades) provide a starting point by establishing students' baseline knowledge and readiness levels. Informal assessments (e.g., observations, one-on-one conversations, or ongoing formative assessments) allow

Table 4. Sample ChatGPT Prompts—Kiara's Astrophysics Exploration.

Task	Examples
Student scenario prompts	Broad: "What's the most unusual thing about black holes?"
	Specific: "If I fell into a black hole, what would happen?"
	Focused: "Can you explain Hawking radiation?"
	Detailed: "How do scientists study black holes if we can't see them? What tools do they use?"
	Application: "How might understanding black holes help us learn about the beginning of the universe?"
	Reflection: "If you could ask a famous scientist like Neil deGrasse Tyson one question about black holes, what would it be and why?"
Student final comprehensive prompt	"You're a young scientist who just discovered a new type of black hole. Create a report for other student scientists that includes
	- A fun name for your black hole
	- A drawing or description of what it looks like
	- Three special features that make it different from other black holes
	- How it behaves (what it does, how it moves, what it "eats")
	- Two questions you still have about your discovery
	Be as creative and scientific as you can."
Teacher to-dos	1. Use ChatGPT to generate analogies that explain complex concepts in simpler terms
	2. Create a "spot the mistake" game: Intentionally include a few errors in ChatGPT's response for Kiara to identify and correct
	3. Design a multi-sensory activity based on ChatGPT's explanations (e.g., a physical model, a sound representation of black hole behavior)
	4. Use ChatGPT to generate ideas for connecting astrophysics concepts to other subjects Kiara enjoys

teachers to continuously gauge each student's interests, strengths, and areas needing growth.

For example, when Sophia expresses her interest in marine ecosystems, ChatGPT doesn't just provide information. It can help chart a course through this vast field of study based on both her formal and informal assessment data. Sophia's teacher, Mr Edwards, could begin by reviewing formal assessments that show her advanced reading level and academic performance, while informal observations and check-ins reveal her feeling of being unchallenged in school. With this combined insight, Mr. Edwards can use ChatGPT to create an individualized learning plan that adjusts to Sophia's unique needs.

Mr. Edwards could type in to ChatGPT, "Create an individualized learning plan for an advanced 10th grade student who reads on a college level based on her interests (marine biology, deep-sea ecosystems), strengths (needs differentiated learning, feels unchallenged in school), and goals (wants to be a marine biologist and contribute to marine conservation efforts, but has limited access to AP classes)." When this prompt is entered into ChatGPT, it responds with suggested independent study projects and online courses in marine biology, encourages her participation in science clubs and competitions, shares ideas for research resources, and provides conservation project ideas and career planning. This personalized guidance helps Sophia navigate her educational path with confidence and purpose and provides her teachers with support in curriculum planning. It is important, however, to be careful not to use private individual student data or information (e.g., names, test data); schools should develop AI policies and provide training for teachers that include ethical considerations and data privacy.

Setting Goals

One of the key aspects of effective mentoring is helping mentees set and achieve goals. ChatGPT excels at this, assisting gifted learners like Sophia, Mateo, and Kiara in establishing clear, achievable objectives aligned with their capabilities and aspirations. Sophia's long-term goal is to contribute to marine conservation efforts. When prompted with specific questions from Sophia and her human advisers (e.g., caregivers, teachers, counselor), ChatGPT can help break this down into manageable steps. It might suggest short-term goals like mastering the classification of marine species, understanding the basics of oceanography, or completing a project on local marine ecosystems. For each goal, ChatGPT can provide resources, set milestones, and offer encouragement along the way.

Facilitating Reflection on Learning

Self-reflection is a crucial component of effective learning and is included as a goal in the NAGC Standards. Standard 2, *Assessment*, recommends educators use evidence-based practices to, "Educators provide opportunities for students to set personal goals, keep records, and monitor their own learning progress" (National Association for Gifted Children, 2019, p. 9). It is important to remember that teachers still play a vital role in their learning experiences. For example, after Kiara completes a virtual astrophysics course, Mrs. Spencer might ask, "What was the most surprising thing you learned about dark matter? How has this new knowledge changed your understanding of astrophysics?" These questions encourage metacognition by having Kiara pause and consider her learning journey, helping her understand her thought processes, identify areas for improvement, and recognize her achievements. This activity aligns with NAGC Standard 3 Evidence-Based Practice 3.5.1, "Educators model and teach metacognitive models to meet the needs of students with gifts and talents such as self-assessment, goal setting, and monitoring of learning" (National Association for Gifted Children, 2019, p. 11). Through this guided reflection, Kiara not only consolidates her knowledge but also develops metacognitive skills that will serve her well in future academic pursuits. She learns to analyze her own learning strategies, identify what works best for her, and adjust her approach accordingly. Her teacher can ask her to write down her reflection responses in a journal and then use them to facilitate a check-in one-on-one session discussing her progress.[1]

Conclusion

Leveraging the power of AI, this innovative approach aims to enhance the mentorship experience by offering personalized guidance, facilitating independent research, providing real-time feedback, supporting career and future planning, and creating robust, interactive learning opportunities. Throughout this investigation, ChatGPT has demonstrated effectiveness in tailoring educational content, providing consistent virtual mentorship, and fostering reflective learning practices. By aligning with the NAGC standards, ChatGPT ensures it meets the unique cognitive and affective needs of gifted students.

Educators are encouraged to explore and integrate ChatGPT into educational practices to effectively support gifted learners. The integration of AI tools like ChatGPT has the potential to transform traditional mentorship models, making them more dynamic and responsive to individual student needs. By merging human expertise with AI capabilities, a more enriching and supportive learning environment can be cultivated, nurturing well-rounded, future-ready leaders. AI will not replace the need for highly qualified teachers, yet it significantly enhances educators' ability to mentor and guide students. Educators are urged to explore the use of ChatGPT and other AI tools to elevate teaching strategies and better support students' growth and development. Together, this approach fosters a new era of education that empowers gifted learners to reach their full potential.

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Note

1. The GenAI tool Claude 3.5 was used to help create sample activities for the student case scenarios in this manuscript.

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Bios

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